

energy processes can still be established implementing specific symmetry properties, such as the MFV hypothesis (Sect. IV). The experimental tests of such correlations allow us to test/establish general features of the new theory which holds independently of the dynamical details of the model. In particular, B , D and K decays are extremely useful in determining the flavor-symmetry breaking pattern of the NP model.

A. Bounds from $\Delta F = 2$ down-type transitions

The starting points for this analysis is the observation that in several realistic NP models we can neglect non-standard effects in all cases where the corresponding effective operator is generated at the tree-level within the SM. This general assumption implies that the experimental determination of the CKM matrix via tree-level processes is free from the contamination of NP contributions. Using this determination we can unambiguously predict meson-antimeson mixing and FCNC amplitudes within the SM and compare it with data, constraining the couplings of the $\Delta F = 2$ operators in (3.1). Each $\Delta F = 2$ amplitude is then conveniently parametrized in terms of the shift induced in the modulo and the CPV phase, or the real and imaginary part [10, 11]:

$$\frac{\langle B_q | \mathcal{L}_{\text{eff}} | \overline{B}_q \rangle}{\langle B_q | \mathcal{L}_{\text{SM}} | \overline{B}_q \rangle} = C_{B_q} e^{2i\phi_{B_q}} , \quad \frac{\text{Re}[\langle K^0 | \mathcal{L}_{\text{eff}} | \overline{K}^0 \rangle]}{\text{Re}[\langle K^0 | \mathcal{L}_{\text{SM}} | \overline{K}^0 \rangle]} = C_{\Delta m_K} \xrightarrow{\text{Re} \rightarrow \text{Im}} C_{\epsilon_K} , \quad (3.2)$$

An updated analysis of these constraints has been presented in [12] (see Fig. 2). The main conclusions that can be drawn from this analysis can be summarized as follows:

(i) In all the three accessible short-distance amplitudes ($K^0 - \overline{K}^0$, $B_d - \overline{B}_d$, and $B_s - \overline{B}_s$) the magnitude of the new-physics amplitude cannot exceed, in size, the SM short-distance contribution. The latter is suppressed by both the GIM mechanism and the hierarchical structure of the CKM matrix,

$$\mathcal{A}_{\text{SM}}^{\Delta F=2} \approx \frac{G_F^2 m_t^2}{16\pi^2} (V_{ti}^* V_{tj})^2 \times \langle \overline{M} | (\overline{Q}_{Li} \gamma^\mu Q_{Lj})^2 | M \rangle \times F \left(\frac{M_W^2}{m_t^2} \right) , \quad (3.3)$$

where F is a loop function of $\mathcal{O}(1)$. As a result, new-physics models with TeV-scale flavored degrees of freedom and $\mathcal{O}(1)$ effective flavor-mixing couplings are ruled out. To set explicit bounds, let us consider for instance the subset of left-handed $\Delta F = 2$ operators in the generic effective Lagrangian in (3.1), namely

$$\Delta \mathcal{L}^{\Delta F=2} = \sum_{i \neq j} \frac{c_{ij}}{\Lambda^2} (\overline{Q}_{Li} \gamma^\mu Q_{Lj})^2 , \quad (3.4)$$